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Caesar Cipher & Playfair Cipher Encryption Decryption Technique

Method · January 2023

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Experiment Name: Perform encryption, decryption using the following substitution techniques: Ceaser Cipher & Playfair Cipher.

Objectives:

- ✓ To encrypt and decrypt the given message by using Ceaser Cipher encryption algorithm.
- ✓ To implement a program to encrypt a plain text and decrypt a cipher text using play fair Cipher substitution technique.

Theory:

Caesar Cipher Technique:

The Caesar Cipher technique is one of the earliest and simplest methods of encryption technique. It's simply a type of substitution cipher, i.e., each letter of a given text is replaced by a letter with a fixed number of positions down the alphabet. For example, with a shift of 1, A would be replaced by B, B would become C, and so on. The method is apparently named after Julius Caesar, who apparently used it to communicate with his officials.

Thus, to cipher a given text we need an integer value, known as a shift which indicates the number of positions each letter of the text has been moved down. The encryption can be represented using modular arithmetic by first transforming the letters into numbers, according to the scheme, A = 0, B = 1, …, Z = 25. Encryption of a letter by a shift n can be described mathematically as.

$$\mathbf{E_n(n) = (x + n) \bmod 26}$$

(Encryption Phase with shift n)

$$\mathbf{D_n(n) = (x - n) \bmod 26}$$

(Decryption Phase with shift n)

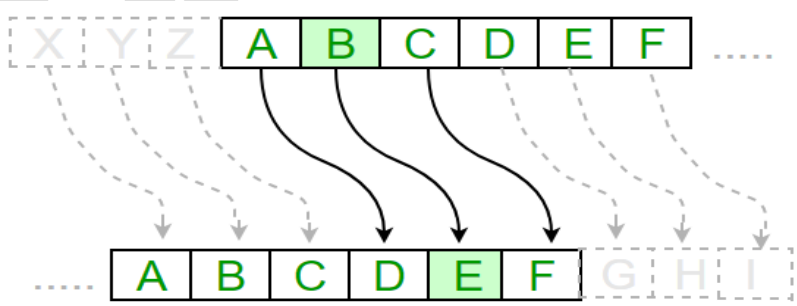

Figure 1: Caesar Cipher Technique.

Examples:

Text: ABCDEFGHIJKLMNOPQRSTUVWXYZ

Shift: 23

Cipher: XYZABCDEFGHIJKLMNOPQRSTUVW

Text: ATTACKATONCE

Shift: 4

Cipher: EXXEGOEXSRGI

Algorithms:

1. In Ceaser Cipher each letter in the plaintext is replaced by a letter some fixed number of positions down the alphabet.
2. For example, with a **left shift of 3**, **D** would be replaced by **A**, **E** would become **B**, and so on.
3. The encryption can also be represented using modular arithmetic by first transforming the letters into numbers, according to the scheme, **A = 0, B = 1, Z = 25.**
4. Encryption of a letter x by a shift n can be described mathematically as,

$$En(x) = (x + n) \bmod 26$$

5. Decryption is performed similarly,

$$Dn(x) = (x - n) \bmod 26$$

Program:

CaesarCipher.java

```
class caesarCipher {
    public static String encode(String enc, int offset) {
        offset = offset % 26 + 26;
        StringBuilder encoded = new StringBuilder();
        for (char i : enc.toCharArray()) {
            if (Character.isLetter(i)) {
                if (Character.isUpperCase(i)) {
                    encoded.append((char) ('A' + (i - 'A' + offset) % 26));
                } else {
                    encoded.append((char) ('a' + (i - 'a' + offset) % 26));
                }
            } else {
                encoded.append(i);
            }
        }
        return encoded.toString();
    }
    public static String decode(String enc, int offset) {
```

```
        return encode(enc, 26 - offset);
    }
    public static void main(String[] args) throws java.lang.Exception {
        String msg = "International Islamic University Chittagong";
        System.out.println("Simulating Caesar Cipher\n------------------------");
        System.out.println("Input: " + msg);
        System.out.printf("Encrypted Message: ");
        System.out.println(caesarCipher.encode(msg, 3));
        System.out.printf("Decrypted Message: ");
        System.out.println(caesarCipher.decode(caesarCipher.encode(msg, 3), 3));
    }
}
```

Output:

```
Simulating Caesar Cipher
------------------------
Input: International Islamic University Chittagong
Encrypted Message: Lqwhuqdwlrqdo Lvodplf Xqlyhuvlwb Fklwwdjrqj
Decrypted Message: International Islamic University Chittagong
```

<u>Playfair Cipher Encryption Technique:</u>

The Playfair cipher encryption technique can be used to encrypt or encode a message. It operates exactly like typical encryption. The only difference is that it encrypts a digraph, or a pair of two letters, as opposed to a single letter.

An initial 5×5 matrix key table is created. The plaintext encryption key is made out of the matrix's alphabetic characters. Be mindful that you shouldn't repeat the letters. There are 26 alphabets, however, there are only 25 spaces in which we can place a letter. The matrix will delete the extra letter because there is an excess of one letter (typically J). Despite this, J is there in the plaintext before being changed to I.

The Playfair Cipher Encryption Algorithm:

The Algorithm consists of 2 steps:

Generate the key Square (5×5): The key square is a 5×5 grid of alphabets that acts as the key for encrypting the plaintext. Each of the 25 alphabets must be unique and one letter of the alphabet (usually J) is omitted from the table (as the table can hold only 25 alphabets). If the plaintext contains J, then it is replaced by I.

The initial alphabets in the key square are the unique alphabets of the key in the order in which they appear followed by the remaining letters of the alphabet in order.

Algorithm to encrypt the plain text: The plaintext is split into pairs of two letters (digraphs). If there is an odd number of letters, a Z is added to the last letter.

For example:

PlainText: "instruments"

After Split: 'in' 'st' 'ru' 'me' 'nt' 'sz'

1. Pair cannot be made with same letter. Break the letter in single and add a bogus letter to the previous letter.

   Plain Text: "hello"

   After Split: 'he' 'lx' 'lo'

   Here 'x' is the bogus letter.

2. If the letter is standing alone in the process of pairing, then add an extra bogus letter with the alone letter

   Plain Text: "helloe"

   After Split: 'he' 'lx' 'lo' 'ez'

   Here 'z' is the bogus letter.

<u>Rules for Playfair Cipher Encryption</u>

Rule 1: If both the letters are in the same column: Take the letter below each one (going back to the top if at the bottom).

For example:

Diagraph: "me"

Encrypted Text: cl

Encryption:

m -> c

e -> l

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Rule 2: If both the letters are in the same row: Take the letter to the right of each one (going back to the leftmost if at the rightmost position).

For example:

Diagraph: "st"

Encrypted Text: tl

Encryption:

s -> t

t -> l

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Rule 3: If neither of the above rules is true: Form a rectangle with the two letters and take the letters on the horizontal opposite corner of the rectangle.

For example:

Diagraph: "nt"

Encrypted Text: rq

Encryption:

n -> r

t -> q

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Algorithm:

mhk@iit

1. To encrypt a message, one would break the message into diagrams (groups of 2 letters)
2. For example, "HelloWorld" becomes "HE LL OW OR LD".
3. These diagrams will be substituted using the key table.
4. Since encryption requires pairs of letters, messages with an odd number of characters usually append an uncommon letter, such as "X", to complete the final diagram.
5. The two letters of the diagram are considered opposite corners of a rectangle in the key table. To perform the substitution, apply the following 4 rules, in order, to each pair of letters in the plaintext:

Program:

playfairCipher.java

```
import java.awt.Point;
class playfairCipher {
    private static char[][] charTable;
    private static Point[] positions;
    private static String prepareText(String s, boolean chgJtoI) {
        s = s.toUpperCase().replaceAll("[^A-Z]", "");
        return chgJtoI ? s.replace("J", "I") : s.replace("Q", "");
    }
    private static void createTbl(String key, boolean chgJtoI) {
        charTable = new char[5][5];
        positions = new Point[26];
        String s = prepareText(key + "ABCDEFGHIJKLMNOPQRSTUVWXYZ", chgJtoI);
        int len = s.length();
        for (int i = 0, k = 0; i < len; i++) {
            char c = s.charAt(i);

            if (positions[c - 'A'] == null) {
                charTable[k / 5][k % 5] = c;
                positions[c - 'A'] = new Point(k % 5, k / 5);
                k++;
            }
        }
    }
    private static String codec(StringBuilder txt, int dir) {
        int len = txt.length();
        for (int i = 0; i < len; i += 2) {
            char a = txt.charAt(i);
            char b = txt.charAt(i + 1);
```

```
            int row1 = positions[a - 'A'].y;
            int row2 = positions[b - 'A'].y;
            int col1 = positions[a - 'A'].x;
            int col2 = positions[b - 'A'].x;
            if (row1 == row2) {
                col1 = (col1 + dir) % 5;
                col2 = (col2 + dir) % 5;
            } else if (col1 == col2) {
                row1 = (row1 + dir) % 5;
                row2 = (row2 + dir) % 5;
            } else {
                int tmp = col1;
                col1 = col2;
                col2 = tmp;
            }
            txt.setCharAt(i, charTable[row1][col1]);
            txt.setCharAt(i + 1, charTable[row2][col2]);
        }
        return txt.toString();
    }
    private static String encode(String s) {
        StringBuilder sb = new StringBuilder(s);
        for (int i = 0; i < sb.length(); i += 2) {
            if (i == sb.length() - 1) {
                sb.append(sb.length() % 2 == 1 ? 'X' : "");
            } else if (sb.charAt(i) == sb.charAt(i + 1)) {
                sb.insert(i + 1, 'X');
            }
        }
        return codec(sb, 1);
    }
    private static String decode(String s) {
        return codec(new StringBuilder(s), 4);
    }
    public static void main(String[] args) throws java.lang.Exception {
        String key = "MONARCHY";
        String txt = "Cryptography and Network Security"; /* make sure string length is
even */ /* change J to I */
        boolean chgJtoI = true;
```

```
        createTbl(key, chgJtoI);
        String enc = encode(prepareText(txt, chgJtoI));
        System.out.println("Simulating Playfair Cipher\n----------------------");
        System.out.println("Input Message : " + txt);
        System.out.println("Encrypted Message : " + enc);
        System.out.println("Decrypted Message : " + decode(enc));
    }
}
```

Output:

```
Simulating Playfair Cipher
----------------------
Input Message: Cryptography and Network Security
Encrypted Message: DMHQPRKNOSYBRAYRKLVNDTLIEMAKQD
Decrypted Message: CRYPTOGRAPHYANDNETWORKSECURITY
```

<u>Lab Task 1:</u> Rewrite a Caesar Cipher Description Technique Program by using the Shifting key "A". Assume "A" is equal to the last two digits of your student ID.

<u>Lab Task 2:</u> Rewrite a Playfair Cipher Description Technique Program using the String key "X". Assume "X" is equal to any word with at least six non-repetition alphabet letters.

<u>Conclusion:</u>

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Question Answer:

1. What is the main weakness of Cesar cipher?
2. Why is the Caesar cipher not secure?
3. Can Caesar cipher be cracked?
4. How is Caesar cipher broken?
5. What makes a cipher weak?
6. Is Playfair cipher breakable?
7. What is the weakness of Playfair cipher?
8. What is the alternative name of Playfair cipher?
9. Why J is omitted in Playfair cipher?
10. How do you decode a Playfair cipher?

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